

EFFECTS OF A 10-WEEK RESISTANCE EXERCISE PROGRAM ON SOCCER KICK BIOMECHANICS AND MUSCLE STRENGTH

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ABSTRACT

Manolopoulos, E, Katis, A, Manolopoulos, K, Kalapotharakos, V, and Kellis, E. Effects of a 10-week resistance exercise program on soccer kick biomechanics and muscle strength. *J Strength Cond Res* 27(12): 3391–3401, 2013—The purpose of the study was to examine the effects of a resistance exercise program on soccer kick biomechanics. Twenty male amateur soccer players were divided in the experimental group (EG) and the control group (CG), each consisting of 10 players. The EG followed a 10-week resistance exercise program mainly for the lower limb muscles. Maximal instep kick kinematics, electromyography, and ground reaction forces (GRFs) as well as maximum isometric leg strength were recorded before and after training. A 2-way analysis of variance showed significantly higher ball speed values only for the EG ($26.14 \pm 1.17 \text{ m} \cdot \text{s}^{-1}$ vs. $27.59 \pm 1.49 \text{ m} \cdot \text{s}^{-1}$ before and after training, respectively), whereas no significant differences were observed for the CG. The EG showed a decline in joint angular velocities and an increase in biceps femoris electromyography of the swinging leg during the backswing phase followed by a significant increase in segmental and joint velocities and muscle activation of the same leg during the forward swing phase ($p < 0.05$). The EG also showed significantly higher vertical GRFs and rectus femoris and gastrocnemius activation of the support leg ($p < 0.05$). Similarly, maximum and explosive isometric force significantly increased after training only for the EG ($p < 0.05$). These results suggest that increases in soccer kicking performance after a 10-week resistance training program were accompanied by increases in maximum strength and an altered soccer kick movement pattern, characterized by a more explosive backward-

forward swinging movement and higher muscle activation during the final kicking phase.

KEY WORDS weight training, soccer, ball speed, strength

INTRODUCTION

One of the most important aims of soccer training was to improve soccer kick performance because it is the most important technical skill used by players to score a goal (14). Maintaining high kicking performance during the 90-minute game requires high levels of muscle strength, power, endurance, and technique (28). Therefore, different types of training can be applied to improve kick performance.

Improvement of soccer kick performance requires understanding of the basic determinants of the kick (14,17). A “faster” soccer kick may be associated with a higher velocity of the foot at ball impact and a higher angular velocity of the knee joint and a faster approach of the player to the ball (14). Improvement of these aspects of the kick may be achieved by designing appropriate soccer training programs.

Muscle strength largely contributes to the final speed of the ball at impact (15,20,29,32). In particular, significant positive correlations between knee extensor (2,4,20,33,34) and hip flexor (4,9,23) muscle strength and kicking performance have been reported ($r = 0.68$ – 0.91), although others found no association (1,5,21,22). Nevertheless, studies reported that strengthening programs using different types of exercises (isometric, plyometric, and simulation training) improve soccer kick performance (3,9,29,30,31,33,38) (see 39 for a review). Improvements in ball speed of the kick were observed after a training program consisting of soccer drills and movements (33), 12 weeks of strength and power training (38), 5 weeks of electrostimulation (3), 6 weeks of isokinetic training (9), and 12 weeks (30) and 10 weeks (31) of plyometric training. These improvements were accompanied by enhancement of jumping and sprinting performance (29,38), isometric (3) and explosive muscle strength (30,31), and soccer kick accuracy (9). A limitation of the aforementioned studies is that they examined

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adaptations in final soccer performance, although limited information is provided on how improvements in various components of physical conditioning are transferred into kicking technique and performance. Therefore, it is not clear what aspects of actual game performance improve after training.

Very few studies examined the effects of training interventions on soccer kick biomechanics (20,27). Perez-Gomez et al. (27) found increased peak angular velocity of the knee during kicking after 6 weeks of combined strength training and plyometric training. Manolopoulos et al. (20) reported that 10 weeks of soccer-specific exercises significantly improved maximum ball speed and the linear velocity of the foot and the angular velocity of all joints during the final phase of the kick. This was achieved by a faster transition from the backward to the forward swinging phase, which indicates a better co-ordination after soccer-specific exercises. None of these studies have applied resistance training programs and therefore the effects of strength training on soccer kick biomechanics are not clear. One may assume that strength training exercises may improve powerful kick performance by increasing muscle strength, but the mechanism of such improvement is not clear. In a recent review, Young and Rath (39) concluded that there is no sufficient evidence on the characteristics of strength training programs that enhance power kick performance.

Since kicking is a multi-articular task, an increase in final ball speed may be due to increases in absolute muscle strength capacity but it may also be due to altered activation patterns during the kick, as a result of training. Previous studies reported that the level and the timing of the activity of several muscles of the swinging leg largely affects ball speed (8,32). Particularly, hip flexors are activated first, followed by the rectus femoris (RF) and finally by vastus lateralis (32). In addition, muscle activation of the support leg provides the necessary stability so that the whole body rotates and quickly swings the kicking leg (16,23). Although it is clear that strength training improves soccer kick performance, its effects on muscle activation levels during soccer kick are not clear. Some evidence indicates that improvements in soccer kick speed are accompanied by an increase in RF activation while other lower limb muscles activity remains unaltered (20). In fact, Manolopoulos et al. (20) concluded that any changes in soccer kick performance after soccer-specific training are not because of general alterations in muscle activation. It is not clear, however, that strength training adaptations would be similar to soccer-specific training ones. Examination of muscle activation patterns may provide more useful information on how any gains in absolute muscle strength are transferred into skill performance.

It is apparent that although many studies have examined soccer player adaptations after various training interventions, a few studies examined the biomechanics of soccer kick after resistance training. Furthermore, although the use of sport-specific exercises as part of soccer training is frequently advocated (39), resistance training exercises are part of reg-

ular soccer training practice, especially in the pre-season period. In training practice, soccer coaches need to know how strength training exercises improve soccer kick performance. Therefore, in this study, we aimed to examine the effects of a typical strength training program on kinematics, kinetics, and electromyographic characteristics of fast soccer kicks. The main hypotheses tested were that resistance exercise would (a) increase ball speed and maximum joint and segmental speeds, (b) affect muscle activation, and (c) affect ground reaction forces (GRFs) during the kick.

METHODS

Experimental Approach to the Problem

To test our hypotheses, a randomized group pre-post test design has been used. The independent variable was the resistance training program. The dependent variables were the velocity of the ball, GRFs, and kinematic and electromyographic characteristics of kicking performance. Each variable was measured on 2 occasions: 1 week before the start and 1 week after the end of the training program.

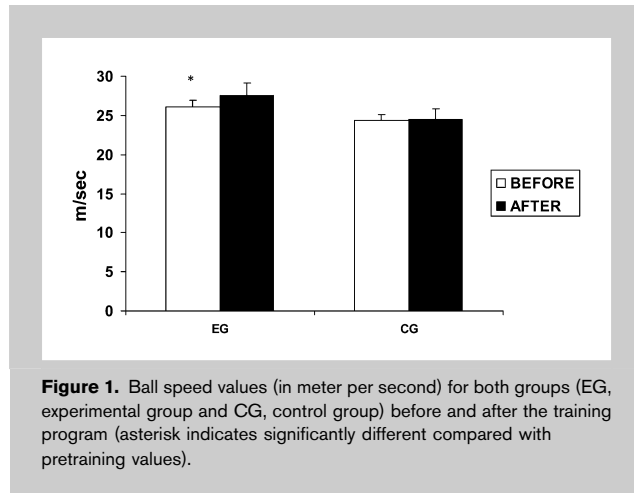
Subjects

Twenty healthy amateur soccer players volunteered to participate in the present study. All participants played for the same team and trained at least 3 times (regular soccer training, including resistance exercises, where appropriate) plus a game per week for the last 2 years before testing. All participants signed informed consent forms approved by the University Ethics Committee. All participants passed medical examination within 15 days before the training program and had no injury of their lower limbs. Criteria for exclusion from this study included that players had fewer than 5 years of systematic participation in soccer, they experienced an orthopedic injury in the last 6 months, and an injury during the training process. All participants were familiarized with resistance training exercises; however, they did not systematically participate in other resistance or strength training programs during the intervention period. During the strength training period, participants followed their regular sleep and nutritional program while they were informed that they could drop out at any time.

Participants were randomly allocated in 2 groups based on age, mass, height, and training age. The experimental group (EG) consisted of 10 players (age: 19.9 ± 1.4 years, mass: 74.8 ± 9.1 kg, height: 177.4 ± 6.7 cm, training age: 5.5 ± 1.2 years) and the control group (CG) consisted of 10 players (age: 20.6 ± 1.3 years, mass: 71.5 ± 6.7 kg, height: 175.2 ± 3.4 cm, training age: 6.1 ± 1.4 years). Seven players from the EG and 7 players from the CG preferred to kick the ball with the right foot.

Procedures

Training Program. In addition to regular training practice, the EG followed a 10-week conventional resistance exercise program, 3 times per week. The training program was applied during the pre-season, and each session was performed at same time of the day.



Before the training program, the 1 repetition maximum (RM) was determined for all resistance training exercises, with the exception of simulated kicking actions. The training program was based on previous guidelines (10) and it is presented in detail in Appendix 1. During the first 2 weeks, the aim was to improve general strength of the upper and the lower limb muscles. For this reason, 10 exercises of various muscle groups together with simulated kicking actions against resistance provided by a rubber band attached at the ankle of the swinging leg were performed. Three sets of 10 repetitions for each exercise at 50% of RM were performed. The next 3 weeks, the program included performance of 8 exercises at relatively low intensity (submaximal) loads (70% RM, 3 sets $\times$ 10 repetitions). For the first 5 weeks, a 60-second rest interval between exercises and a 3-minute rest interval between sets were provided.

During the sixth, seventh, and eighth week, the players performed 6 exercises at a higher intensity compared with first 5 weeks (see Appendix 1). For each exercise, 5 sets were performed by progressively reducing the number of repetitions

(from 6 to 2) and simultaneously increasing the load (from 70 to 90% RM). During the last 2 weeks, training intensity increased further with a range of 80 to 95% RM. For each exercise, 4 sets were performed by progressively reducing the number of repetitions (from 4 to 1) and simultaneously increasing the load (from 80 to 95% RM). For weeks 6 to 10, a 90-second rest interval between exercises and a 3-minute rest interval between sets were provided.

The CG performed regular soccer-specific training, but they did not participate in systematic strength training programs during the main experimental period.

Instep Soccer Kick. A warm-up consisting of 10-minute submaximal running on a treadmill, stretching exercises, mainly for the lower limbs, and several practice kicks was conducted. Kicks were performed against the center of a goalpost (3.5 m height $\times$ 2.5 m width) located 6 m from the ball. A standard sized (size 5) and inflated ball was used for the measurements, although participants wore soccer shoes.

Each participant performed 5 consecutive kicking trials after a 2-step straight approach run. Particularly, the participant was standing 2 m behind the ball and then performed first, a step with the right leg to the kicking direction (for a right kicking player) and afterwards a second step with the support leg on the force plate and kicked the ball with the swinging leg. A 30-second rest interval between consecutive kicks was provided. Participants were instructed to kick the ball as hard as possible without accuracy demands. If a kick was not performed according to investigator's guidelines, the kick was repeated. The trial which displayed the highest ball speed was further analyzed. This is because our aim was to examine maximal player performance after training, which is best represented by the most powerful kick (the kick which displays the highest ball speed). The same procedure was followed in the second measurements, after the completion of the training program. Pretest and posttest measurements were made at the same

time of the day to ensure consistent testing (environmental and time) conditions before and after training.

TABLE 1. Mean ($\pm$ SD) phase temporal parameters (in milliseconds) and vertical GRFs (in Newton) during the kicking trials for both groups before and after the training program.*

Variables	Experimental group		Control group	
	Before	After	Before	After
Second phase duration	65 $\pm$ 13	57 $\pm$ 12	66 $\pm$ 15	65 $\pm$ 11
Third phase duration	73 $\pm$ 11	70 $\pm$ 10	75 $\pm$ 9	76 $\pm$ 9
Total duration	138 $\pm$ 12	127 $\pm$ 9	141 $\pm$ 11	141 $\pm$ 14
GRF _{vertical}	2067 $\pm$ 352	2219 $\pm$ 412†	1562 $\pm$ 412‡	1622 $\pm$ 293‡

*GRF = ground reaction force.

†Significantly different compared with the pretraining values at $p < 0.05$.

‡Significantly different compared with corresponding experimental group value at $p < 0.05$.

Kinematic Characteristics. Kinematic data were collected using an Ariel Performance Analysis system (Ariel Dynamics, Inc., San Diego, CA, USA). Two high-speed digital video cameras (JVC 9800), at a frame rate of 120 Hz with a high-speed shutter, were used to record the kicking motion. The cameras were placed on tripods at a height of 1.4 m. One camera was placed 6 m behind and 8

TABLE 2. Mean ($\pm$ SD) joint linear velocity of the swinging leg (in meter per second) for both groups before and after the training program.*

Variables	Experimental group		Control group	
	Before	After	Before	After
First phase				
LVfoot	7.65 $\pm$ 0.59	6.89 $\pm$ 1.66	7.43 $\pm$ 1.61	7.46 $\pm$ 1.54
LVankle	6.32 $\pm$ 0.92	5.89 $\pm$ 1.05	6.76 $\pm$ 1.06	6.75 $\pm$ 1.02
LVknee	4.43 $\pm$ 0.77	4.71 $\pm$ 1.15	3.88 $\pm$ 1.19	3.99 $\pm$ 1.28
LVhip	3.43 $\pm$ 0.85	3.14 $\pm$ 0.68	3.77 $\pm$ 0.97	3.81 $\pm$ 1.03
Second phase				
LVfoot	8.43 $\pm$ 3.84	11.01 $\pm$ 4.47	12.73 $\pm$ 1.82†	11.61 $\pm$ 3.66
LVankle	6.40 $\pm$ 3.34	9.20 $\pm$ 2.86	8.20 $\pm$ 3.62†	8.95 $\pm$ 3.10
LVknee	8.46 $\pm$ 2.01	9.19 $\pm$ 1.61	9.40 $\pm$ 2.39†	9.32 $\pm$ 2.02
LVhip	3.14 $\pm$ 0.80	4.48 $\pm$ 2.03‡	3.30 $\pm$ 1.02	3.69 $\pm$ 0.96
Third phase				
LVfoot	20.95 $\pm$ 2.71	23.21 $\pm$ 3.01‡	20.38 $\pm$ 3.39	20.44 $\pm$ 3.30†
LVankle	18.26 $\pm$ 2.46	20.44 $\pm$ 2.70‡	17.37 $\pm$ 2.22	17.19 $\pm$ 2.14†
LVknee	5.43 $\pm$ 1.49	6.87 $\pm$ 1.19‡	5.59 $\pm$ 1.40	5.62 $\pm$ 1.59
LVhip	4.20 $\pm$ 2.08	4.71 $\pm$ 1.15	3.71 $\pm$ 1.98	3.96 $\pm$ 1.92

*LV = linear velocity.

†Significantly different compared with corresponding experimental group value at $p < 0.05$.‡Significantly different compared with the pre-training values at $p < 0.05$.

m to the right of the ball and the other 8 m in front and 9 m to the right. The optical axis between the 2 cameras was 90°.

Reflection markers were placed on selected anatomical locations on both sides to identify segments and joints: the

right-handed orthogonal reference frame fixed on the ground with the Y axis pointing toward the direction of the ball, the Z axis vertically upward, and the X axis perpendicular to Y and Z axes. The resulting displacement-time data of each marker

were filtered using a second-order Butterworth digital filter with zero-order phase lag. Optimal cut-off frequencies were chosen by comparing the residuals of the difference between filtered and unfiltered signals and ranged from 6 to 12 Hz. To avoid any joint data distortion because of impact, a special procedure was followed. Particularly, ball impact point was specified within each data set. The curves were then smoothed to obtain a more close fit of the raw data curves over a specified small interval (around impact) of the sequence. Adjustment of the curves was achieved by permitting higher frequency variations in raw values. This resulted in smoothed curves that more closely fit the abrupt change in the raw data curves because of ball impact. From

TABLE 3. Mean ($\pm$ SD) joint angular velocity of the swinging leg (in radians per second) of both groups before and after the training program.

Variables	Experimental group		Control group	
	Before	After	Before	After
First phase				
Ankle	-12.75 $\pm$ 2.14	-8.76 $\pm$ 0.81*	-8.42 $\pm$ 4.44†	-8.77 $\pm$ 3.16
Knee	-11.16 $\pm$ 2.28	-8.17 $\pm$ 2.78*	-9.66 $\pm$ 2.58	-10.39 $\pm$ 2.58
Hip	-6.18 $\pm$ 2.65	-5.63 $\pm$ 2.72	-6.01 $\pm$ 2.22	-6.04 $\pm$ 2.24
Second phase				
Ankle	-21.60 $\pm$ 12.77	-10.29 $\pm$ 14.25	-14.45 $\pm$ 4.29†	-13.75 $\pm$ 6.33†
Knee	-18.88 $\pm$ 4.86	-14.60 $\pm$ 4.27*	-17.31 $\pm$ 4.57	-16.92 $\pm$ 4.29†
Hip	-10.69 $\pm$ 5.18	-7.38 $\pm$ 2.98*	-11.60 $\pm$ 4.99	-10.47 $\pm$ 5.63†
Third phase				
Ankle	29.68 $\pm$ 4.81	34.18 $\pm$ 5.02*	23.46 $\pm$ 3.86†	25.77 $\pm$ 4.60†
Knee	33.73 $\pm$ 8.21	36.42 $\pm$ 7.88*	24.81 $\pm$ 9.28†	24.59 $\pm$ 8.34†
Hip	6.42 $\pm$ 4.74	7.69 $\pm$ 4.12*	5.32 $\pm$ 3.22	5.69 $\pm$ 3.13

*Significantly different compared with the pretraining values at $p < 0.05$.†Significantly different compared with corresponding experimental group value at $p < 0.05$.

TABLE 4. Mean ($\pm$ SD) normalized electromyographic values (%) of the support and the swinging leg for both groups before and after the training program.*

Variables	Experimental group		Control group	
	Before	After	Before	After
Support leg—phase 1				
RF	36.8 $\pm$ 19.3	44.5 $\pm$ 17.5	46.7 $\pm$ 12.5	45.6 $\pm$ 13.4
GAS	70.3 $\pm$ 15.2	69.0 $\pm$ 23.4	59.9 $\pm$ 9.4	62.7 $\pm$ 6.7
BF	50.3 $\pm$ 18.4	47.0 $\pm$ 13.9	60.9 $\pm$ 19.5	51.4 $\pm$ 11.0
Support leg—phase 2				
RF	53.0 $\pm$ 12.3	72.7 $\pm$ 22.7†	55.8 $\pm$ 10.6	64.6 $\pm$ 16.8‡
GAS	54.4 $\pm$ 9.0	71.3 $\pm$ 10.0†	51.1 $\pm$ 6.6	54.2 $\pm$ 6.0‡
BF	45.7 $\pm$ 13.4	47.2 $\pm$ 16.7	54.9 $\pm$ 21.4‡	48.6 $\pm$ 9.9
Support leg—phase 3				
RF	48.7 $\pm$ 12.5	64.3 $\pm$ 16.5†	54.5 $\pm$ 6.3‡	53.5 $\pm$ 9.8‡
GAS	55.7 $\pm$ 21.3	70.1 $\pm$ 32.5†	57.3 $\pm$ 19.7	49.1 $\pm$ 9.8‡
BF	53.5 $\pm$ 16.8	48.6 $\pm$ 17.0	67.5 $\pm$ 22.6‡	65.6 $\pm$ 22.5‡
Swinging leg—phase 1				
RF	49.4 $\pm$ 14.1	42.9 $\pm$ 23.4	49.8 $\pm$ 11.9	54.3 $\pm$ 16.7‡
BF	30.8 $\pm$ 13.6	55.4 $\pm$ 19.9†	38.4 $\pm$ 14.7	41.5 $\pm$ 20.8‡
VM	31.1 $\pm$ 11.2	37.3 $\pm$ 32.7	40.2 $\pm$ 16.0	49.5 $\pm$ 7.7‡
Swinging leg—phase 2				
RF	85.4 $\pm$ 16.9	70.0 $\pm$ 27.2	70.7 $\pm$ 18.5	66.3 $\pm$ 25.1
BF	57.3 $\pm$ 15.0	80.7 $\pm$ 24.5†	52.6 $\pm$ 16.1	48.7 $\pm$ 16.4‡
VM	71.2 $\pm$ 24.2	60.2 $\pm$ 26.9	68.1 $\pm$ 15.6	68.9 $\pm$ 15.2
Swinging leg—phase 3				
RF	50.9 $\pm$ 10.2	70.8 $\pm$ 16.3†	48.5 $\pm$ 11.7	51.7 $\pm$ 9.5‡
BF	35.2 $\pm$ 19.2	52.3 $\pm$ 23.3†	40.3 $\pm$ 24.0	36.1 $\pm$ 25.1‡
VM	54.3 $\pm$ 16.1	74.2 $\pm$ 9.2†	55.2 $\pm$ 6.5	51.4 $\pm$ 10.9‡

*RF = rectus femoris; GAS = gastrocnemius; BF = biceps femoris; VM = vastus medialis.

†Significantly different compared with the pretraining values at $p < 0.05$.

‡Significantly different compared with corresponding experimental group value at $p < 0.05$.

the smoothed angular displacement data, the hip, knee, and the ankle angular position data were further analyzed. These data were used to calculate joint angular velocities using numerical differentiation (36).

The absolute magnitude of ball velocity (V_{ball}) was calculated from the values of its vertical and horizontal components (18). Particularly, the horizontal component of the ball velocity was calculated as the first derivative of linear regression lines fitted to their nonfiltered displacements. The vertical component was calculated as the first derivative of a quadratic regression line with its second derivative set equal to $-9.81 \text{ m} \cdot \text{s}^{-2}$ fitted to its nonfiltered displacement in the available frames. The velocity of the center of mass of the foot (V_{foot}) was measured after numerical differentiation of the resulting displacement-time data of each marker. The center of mass location of the foot was defined by the toe and heel markers (24). Subsequently, the velocity of V_{foot} at impact was also recorded and was used to calculate the $V_{\text{ball}}/V_{\text{foot}}$ ratio. Kinematic and electromyographic (EMG) data were recorded before start of the kick and ended when the kick was concluded.

Ground Reaction Forces. The vertical, anteroposterior, and medio-lateral components of the GRFs were measured at a sampling frequency of 1,000 Hz using a Kistler force plate (Kistler Type 9281C; Kistler Instruments, Winterthur, Switzerland). The force plate was located in the middle and underneath of a 7-m-long and 2.5-m-wide pathway to avoid disorientation of the player by focusing on stepping inside the force plate when kicking. The kicking trials were performed inside the area of the pathway. The force plate signals were A/D converted and recorded using the analogue converter of the Ariel system. Subsequently, they were analyzed simultaneously with the kinematic and the EMG data.

Electromyographic Characteristics. The EMG activity was recorded using an EMG interface module of the ARIEL system (Ariel Dynamics, Inc.), sampling at 1,000 Hz, a common mode rejection ratio greater than 100 db at 50/60 Hz, a measurement bandwidth ranging from 8 to 500 Hz, and a gain of 400 Hz. Bipolar surface EMG electrodes (interelectrode distance = 1 cm)

with a pre-amplifier were placed on the RF, the vastus medialis (VM), and the long head of biceps femoris (BF) of the swinging leg and the RF, BF and the medial head of the gastrocnemius (GAS) of the support leg. These muscles were selected because they have been reported as significant contributors for the kicking performance (8,26,32).

Synchronization. For data synchronization, an electronic switch system was used to identify the start of the movement and ball impact using both video and EMG data. Particularly, the start of the movement triggered a light emitting diode (LED), visible on both cameras. Similarly, an electronic switch placed underneath the ball triggered another LED, indicating ball contact time. Signals of both switches were fed as separate channels and sampled with the EMG data at 1,000 Hz. The ball signal and the LED signal initiated at the start of the movement were then used to identify the actual ball contact time. The video and EMG signals were then matched using cubic splines.

Maximum Isometric Leg Press Strength. Isometric bilateral leg press strength was measured as a general expression of strength adaptations of the players after training, which is reliable and easy to quantify. Isometric measurements are limited to the joint position where strength is exerted but they can allow quantification of rapid force development, which is an important element of performance. In the present study, a leg press machine was used to perform the leg press measurements (AMD Co. Ltd LC4204-K600). Force signals were detected by a force transducer (sampling rate: 1,000 Hz) and then amplified by a charge amplifier (Analog Devices Module, SB40). The dynamometer was calibrated using known weights before each testing session.

Maximal isometric force and force-time parameters of the bilateral leg extensor muscles (hip, knee, and ankle extensors) were measured in a sitting position, with the knee and hip angles at 110° and 90° , respectively (180° = full extension). During maximum isometric effort, a nonmovable-back chair was supporting the trunk while the subjects had their hands on the dynamometer grips. A Velcro belt was placed around the waist to stabilize the trunk. During testing, there was standardized verbal encouragement to develop the force as quickly as possible and visual feedback of the force-time curve.

Each subject performed 3 maximum voluntary contractions separated by a 3-minute interval. The subjects were instructed to exert maximum effort as fast and hard as possible. The trial in which the highest maximal force was exerted was further analyzed.

Data Analysis

Soccer Kick. To simplify the results, kicking motion was divided into 3 phases (19). Particularly, the first phase was defined from the start of the movement until ground contact. The second phase started from ground contact until the maximum knee flexion angle of the swinging leg. The third phase from maximum knee flexion until initial ball impact.

From the recorded data, the variables examined were the linear displacement and velocity of the thigh, tibia and foot of both legs, the angular displacement and velocity of the hip, knee, and ankle of both legs, the $V_{\text{ball}}/V_{\text{foot}}$ ratio, and the linear velocity of the ball. In addition, the maximum vertical, anteroposterior, and mediolateral GRFs were analyzed.

The EMG data were analyzed using Ariel Performance Analysis Software (APAS; Ariel Dynamics, Inc.). The signals were high-pass filtered with a Butterworth fourth-order zero-lag digital filter at a cut-off frequency of 20 Hz and full-wave rectified. All EMG measurements were normalized by dividing the recorded data by the maximum EMG of each muscle during each kick. Subsequently, the normalized EMG data of each muscle were averaged across each phase of the soccer kick. Consequently, 3 EMG values (phases 1, 2, and 3), for each muscle, were used for further analysis.

Muscle Strength. From the registered force-time curve on the isometric leg press device, maximal force (F_{maxiso}) and explosive force (F_{100}) were determined. Particularly, F_{maxiso} was defined as the highest value of force during each effort. Explosive force (F_{100}) was defined as the amount of force exerted during the first 100 ms of the contraction. F_{maxiso} was divided by subject body mass ($F_{\text{maxiso}}/\text{BW}$) to yield relative strength. The rate of force development ($F_{100}/F_{\text{maxiso}} \times 100$) was also estimated.

Statistical Analyses

The data were checked for normality using the Kolmogorov-Smirnov test and for the homogeneity of variance using the Levenes' test. Two-way analysis of variance (ANOVA) with repeated measures designs were used to examine the differences in each parameter, between the 2 groups, before and after the training period. Significant interactions were followed by applying simple effects and, if significant, post-hoc Tukey's tests were applied to examine significant differences between pairs of means.

The differences between pre-training and posttraining values for each variable were also calculated. The resultant variables were then inserted into a step-wise linear regression analysis and was applied to examine the contribution of angular velocity and GRF adaptations (independent variables) to ball speed (dependent variable) improvements after training. The level of significance was set at $p < 0.05$.

RESULTS

Kinematic Characteristics

Ball Speed. The ANOVA indicated a significant ($p < 0.05$)

TABLE 5. Mean ($\pm$ SD) bilateral strength test parameters for both groups before and after the training program.*

Variable	Experimental group		Control group	
	Before	After	Before	After
F_{maxiso} (N)	2050 $\pm$ 517	2380 $\pm$ 565†	1833 $\pm$ 224‡	1852 $\pm$ 302‡
$F_{\text{maxiso}}/\text{BW}$	2.79 $\pm$ 0.68	3.24 $\pm$ 0.74†	2.61 $\pm$ 0.42	2.64 $\pm$ 0.53‡
F_{100} (N)	1145 $\pm$ 149	1296 $\pm$ 190†	1144 $\pm$ 130	1176 $\pm$ 137‡
RFD ($F_{100}/F_{\text{maxiso}} \times 100$)	55.8 $\pm$ 10.1	54.4 $\pm$ 12.2	62.4 $\pm$ 5.6	63.5 $\pm$ 8.8

* F_{maxiso} = maximum isometric force; $F_{\text{maxiso}}/\text{BW}$ = maximum isometric force divided by body weight; F_{100} = maximum force value during the first 100 ms of the contraction; RFD = rate of force development.

†Significantly different compared with the pre-training values at $p < 0.05$.

‡Significantly different compared with corresponding experimental group value at $p < 0.05$.

interaction (Time $\times$ Group) effect on ball speed values (Figure 1). Post-hoc Tukey's tests indicated that the EG showed significantly higher ball speed values for the EG group compared with pretraining performance ($p < 0.05$). A nonsignificant $V_{\text{ball}}/V_{\text{foot}}$ ratio interaction effect was observed (Table 2, $p > 0.05$).

Temporal Parameters and GRFs. The results of the study showed no significant interaction effects on the duration of the kick (Table 1, $p > 0.05$). In contrast, the ANOVA results indicated a significant interaction effect on vertical GRFs (Table 1, $p < 0.05$). Post-hoc Tukey's tests indicated that the EG significantly increased the vertical GRFs after training. No significant interaction effects were observed on the anteroposterior and the mediolateral GRF values ($p > 0.05$).

Linear Kinematics. There was a statistically significant interaction effect on linear velocity of the hip during the second phase and on the linear velocity of the foot, ankle, and knee during the third phase of the kick (Table 2, $p < 0.05$). Post-hoc Tukey's tests indicated that posttraining linear velocity of the foot, ankle, and knee during the third phase for the EG was significantly higher compared with pretraining values ($p < 0.05$). In addition, group differences were significant for some variables either before or after exercise ($p < 0.05$).

Angular Kinematics. The ANOVAs showed statistically significant interaction effects on joint angular velocities during each phase (Table 3, $p < 0.05$). Post-hoc tests showed that the ankle and knee angular velocity during the first kicking phase significantly decreased for the EG group after training (Table 3, $p < 0.05$). Similarly, the EG showed a decrease in hip and knee angular velocity during the second phase after training. As for the third phase, the EG improved all joints angular velocity after training ($p < 0.05$). No other significant differences in the angular velocity

values were found ($p > 0.05$). In addition, group differences were significant for some variables either before or after exercise ($p < 0.05$).

EMG Characteristics

There was a significant interaction effect ($p < 0.05$) on EMG activation of all swinging leg muscles (Table 4). Post-hoc Tukey's comparisons indicated that the EG showed an increase in BF activation in each kicking phase ($p < 0.05$) and an increase in VM and RF activation during the third kicking phase ($p < 0.05$).

The ANOVA showed that there was a significant interaction effect ($p < 0.05$) on RF and GAS EMG of the support leg (Table 4). Post-hoc Tukey's test comparisons indicated that the EG showed significantly higher GAS (second and third kicking phase) and RF EMG (third phase) after training ($p < 0.05$).

Isometric Leg Press Parameters. The ANOVA followed by a post-hoc Tukey's test comparison showed a significant increase ($p < 0.05$) of F_{maxiso} , $F_{\text{maxiso}}/\text{BW}$, and F_{100} after training (Table 5). A nonsignificant interaction effect on rate of force development was observed (Table 5, $p > 0.05$). In addition, group differences were statistically significant before (F_{maxiso}) and after training (for most variables, $p < 0.05$).

The results of the regression analysis are presented in Table 6. The regression analysis showed a statistically significant prediction ($R^2 = 0.934$, $p < 0.05$) of final ball speed adaptations by a combination of 5 independent variables. Each of these variables displayed a significant correlation with ball speed improvements. The improvements in knee angular velocity at third phase displayed the highest association correlation and association ($R^2 = 0.811$) with ball speed improvements after training. The next steps in regression added the linear angular velocity of the ankle at third phase, the hip angular velocity at first phase, vertical GRF, and F_{maxiso} .

TABLE 6. Percentage changes (mean $\pm$ 95% confidence interval), linear regression coefficient (R^2), and correlation of each included predictor variable with final ball speed.*

	% Change (95% CI)	Correlation R	Regression	
			R^2 change	Standardized (beta) coefficients
Ball speed	5.95 (1.24–10.65)			
AVknee—third phase	13.39 (3.99–22.87)	0.906†	0.811†	0.523
LVanke—third phase	15.68 (6.99–24.37)	0.660†	0.070†	0.298
AVHip—first phase	–22.85 (–48.43 to 3.25)	–0.828†	0.038†	–0.326
MaxvGRF	16.79 (7.40–26.19)	0.070	0.091†	–0.098
F_{maxiso}	15.11 (3.92–29.99)	0.421†	0.059†	0.053
Total R^2			0.935	

*CI = confidence interval; AV = angular velocity; LV = linear velocity; maxvGRF = maximum vertical ground reaction force; F_{maxiso} = maximum isometric force.

†Significant at $p < 0.05$.

DISCUSSION

The main finding of the present study is that the increase in soccer kicking speed after resistance training was accompanied by an increase in maximum speed of lower limb joints, an increase in vertical GRF and in BF and RF activity of the swinging leg, and RF and GAS EMG of the support leg. Based on these findings, the hypotheses of the study are accepted.

A 5.5% increase in ball speed was observed after resistance exercise (Figure 1), which is in line with previous studies after the implementation of strength and power training programs (3,9,29,30,31,33,38). This practically means that soccer players benefited from this training intervention as their ability to kick faster increases the probability to score a goal during a soccer match. The explanation, however, of this increment is complex and should be examined in combination with changes in kick biomechanics and change in maximum strength capacity after training.

The regression analysis (Table 6) confirmed previous suggestions that soccer ball speed is affected by maximal knee and ankle velocity as well as maximum strength (10). As expected, maximum and explosive isometric strength increased by 13–16% after training (Table 5). This is in agreement with previous studies applying different types of strength training programs (3,9,29,30,31,38,39). The key issue is, however, how these strength improvements are transferred into soccer kick performance. Previous studies reported a significant relationship between muscle strength and kicking performance (14,15). It has been suggested that higher muscle strength contributes to higher ball speed in 2 ways: first by allowing lower limb joints to develop higher velocities and, therefore, faster ball speed (20) and second by increasing the effective striking mass, which is considered as being a significant factor for faster kick (14). The results of the present study confirm the first mechanism because a higher linear velocity of the foot and higher linear (Table 2) and angular velocities of all joints at impact (Table 3) were found after training.

Ball speed depends on the velocity of the foot upon impact and the quality of ball and foot impact (14). The higher the speed of the foot before impact, the shorter the foot-ball contact and the higher the ball speed (16,25). For this reason, the ball-to-foot speed ratio has been considered as an index of a successful kick (14,25). In the present study, the ball-to-foot speed ratio was not altered after training (Table 2), probably because both ball and foot speed were altered after training. Within the limitations of this study, this result indicates that soccer kick improvements were not because of changes in quality of foot-ball contact.

It has been found that an increase in the linear velocity of the knee and the angular velocity of the shank results in a higher final foot speed (7), which in turn increases ball speed. Our results indicate that maximal knee angular velocity combined with maximum ankle linear velocity

explains 88.1% of the improvements in soccer kick speed (Table 6). Similar findings were reported after soccer-specific (20) and isokinetic strength (9) training. In addition, we have found that after training, the kick was not accompanied only by a faster final foot speed but also by lower joint angular velocities during the initial and mid-phases of the kick (Table 2). Of these, the reduction in hip angular velocity at the initial phase had a significant contribution to the final soccer ball speed improvement after training (Table 6). This indicates that after training, the players performed the backswing movement through lower segmental and joint velocities followed by a rapid increase in all joint angular velocities as the swinging leg approaches the ball (Table 2). Manolopoulos et al. (20) found even higher reduction in segmental speeds during the preparatory phases followed by higher increases at impact after soccer-specific training exercises.

The reduction in joint speeds during backswing and the increase in the forward swing could be attributed to 2 factors. First, this might be associated with a more effective stretch-shortening cycle of the knee extensors, through a faster transition from a stretching (hip extension and knee flexion) to a shortening phase (hip flexion and knee extension) of the kick (11). This is in line with previous findings and suggestions that the stretch-shortening cycle action of the knee extensors during the kick increases final foot speed and, as a consequence, ball speed (6,26). Second, the higher activation of RF and VM during the last phase of the kick after resistance exercise indicates a higher contribution of these muscles to final knee extension velocity and probably to final performance (Table 4). This supports previous findings, which indicated that final foot speed during the kick is linked with higher hip flexor and knee extensor muscle activation (8).

Although RF and VM activity increased at the final kicking phase, there was a consistent vast increase in BF activation during all kicking phases after training (Table 4). During the initial kicking (backswing) phase, the BF shortens (knee flexed, hip extended) followed by active lengthening (knee extends, hip flexes) during the forward phase. The consistent increase in BF activity coupled with changes in backward (became slower) and forward (became faster) joint speed after training emphasizes the significant role of this muscle during the kicking movement. One might suggest that the increase in posttraining BF EMG may be due to an effort to control hip flexion and knee extension, probably to counteract knee extensor forces (12).

It has been suggested that the support leg also contributes to final kicking speed (16,17). In the present study, the increase in RF and GAS EMG of the support leg (Table 4) during the second and third kicking phases might indicate that the players raise their level of muscle activation to better stabilize the whole leg as it contacts the ground at a faster rate. Furthermore, the EMG increase during the final kicking phase might accompany a small ankle plantar flexion and

knee extension movement of the support leg, which increases momentum transfer to the swinging leg (13).

Collectively, the changes in muscle activation indicate that kicking performance was achieved through an altered co-activation pattern after strength training. The unaltered quadriceps EMG and the increase in BF EMG of the swinging leg (Table 4) may explain the reduction in joint angular velocities during the 2 initial kicking phases (Table 3). In turn, the increase in EMGs of all muscles (Table 3) and the adjustments in support leg activation (Table 4) may partly explain the corresponding increase in segmental velocities at the final kicking phase (Table 3).

Manolopoulos et al. (20) examined the same muscles but found a higher VM activation level of the swinging leg only during the final kicking phase, whereas no changes in support leg EMGs after training were reported. These authors concluded that any changes in soccer kick performance after soccer-specific training are not because of general alterations in muscle activation, which is in contrast to the present results. The main difference between our study and that of Manolopoulos et al. (20) is that they applied soccer-specific training drills as opposed to the resistance training of the present study. Therefore, we can only suggest that increases in kicking speed after strength training are mainly because of enhanced muscle activity and strength, whereas soccer-specific skills achieve a similar outcome through, a relatively, unaltered muscle activation. Further studies are necessary to examine this suggestion.

The resistance training group also showed an increase in vertical GRFs during the kick after training (Table 1). This increase had a minor, but significant, association with soccer ball improvements (Table 6). Higher vertical GRFs are associated with higher acceleration of the center of mass and, therefore, a faster transition of the supported body segments toward the ball (35). In addition, a higher vertical GRF may also indicate an improved stability of the support foot, thus contributing to a more powerful kick performance (17).

The participants of the present study were amateur soccer players, but they trained at least 5 years, 3 times plus a game per week, before the experiment. Therefore, the results of the training program performed in this study are applicable to players with these characteristics. The effects of resistance exercise might differ if the same program is applied in professional players or amateur players without any training experience. Moreover, in the present study, we examined 3 muscles for each leg. A more complete understanding of muscle activation adaptations to training should include examination of more muscles that are active during kicking. Finally, interpretation of EMG patterns is affected by the cross-talk between muscles such that EMG recorded from one area of the muscle is affected by activation of other muscles. Cross-talk is a common issue for all studies examining multiple muscle activation using surface EMGs, especially during multi-articular movements such as the kick. Quantification of this effect during dynamic movements is

difficult, whereas in static movements, the common signal between synergetic muscles can reach 50% (37). In the present study, we took all measures to ensure minimal EMG cross-talk, such as identification of anatomical locations for EMG placement, small interelectrode distance, stabilization of EMG cables, and online inspection of EMG recordings so that all recorded trials are not affected by changes in movement of electrodes.

PRACTICAL APPLICATIONS

Resistance exercise programs are an integral part of strength and conditioning programs in soccer, throughout the season. Although it is clear that such programs improve muscle strength properties, it is not known how muscle strength gains are transferred into actual kicking technique and final performance. Our results show that exercises, such as bar squats, leg press exercises, and simulated kicking exercises, improve powerful kicking performance by raising muscle activation patterns during critical phases of the kick such that the backward phase is slower and the forward swing phase is faster. This mechanism differs compared with previous protocols using soccer-specific exercises, where no changes in muscle activation were observed (20). Based on these results, the use of resistance strength exercises for improving powerful kicking performance is recommended. Coaches may benefit from this type of training during the preseason preparation period when all players return after the summer break season. Furthermore, players who have low strength profiles or their soccer kick is considered by the coach as “less explosive” would also benefit from strength training exercises. Finally, such training is necessary for players who return to training after injury or long absence.

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Appendix 1: The 10-week training program (→ indicates that the next set is performed with less repetitions and a higher load).

	Weeks			
	1–2	3–5	6–8	9–10
Number of exercises	10 + 1	8 + 1	6 + 1	5 + 1
Training exercises	Push-ups, stretching of arms, abdominal, back muscle, hip abduction, hip adduction, knee extension, knee flexion, ankle plantar flexion, ankle dorsiflexion + simulated kicking actions	Hip abduction, hip adduction, knee extension, knee flexion, ankle plantar flexion, ankle dorsiflexion, bar squats, bench press + simulated kicking actions	Knee extension, knee flexion, ankle plantar flexion, ankle dorsiflexion, bench press, bar squats + simulated kicking actions	Knee extension, knee flexion, ankle plantar flexion, bar squats + simulated kicking actions
Overall sets	3	3	5	4
Repetitions	10	10	6 → 5 → 4 → 3 → 2	4 → 3 → 2 → 1
Load	50% of 1 repetition maximum	70% of 1 repetition maximum	70 → 75 → 80 → 85 → 90% of 1 repetition maximum	80 → 85 → 90 → 95% of 1 repetition maximum
Rest	60 seconds between exercises 3 minutes between sets	60 seconds between exercises 3 minutes between sets	90 seconds between exercises 3 minutes between sets	90 seconds between exercises 3 minutes between sets